Mohammad Ibrahim Saleem

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Professional Summary

AI Security Engineer specializing in securing LLM and agentic AI systems across enterprise environments. Experienced in AI threat modeling, adversarial testing, AI governance, and Secure AI SDLC aligned with OWASP Top 10 for LLMs, NIST AI-RMF, and ISO 42001. Hands-on experience performing AI security reviews, building governance workflows, securing MCP-based platforms, and developing privacy-preserving AI data pipelines for production GenAI systems. Skilled in enterprise AI risk assessment, runtime controls, architecture assurance, and secure AI deployment practices.

Education

- **University of Houston** — Houston, TX
  Master of Science in Cybersecurity — *Expected: May 2026*
  – GPA: 3.98/4.0; (Awarded 16k USD in Scholarship) Coursework: Network Security, Secure Enterprise Computing, Cryptography, Data Analysis for Cybersecurity, Cybersecurity Risk Management, Secure Software Design
- **Rajiv Gandhi Proudyogiki Vishwavidyalaya**
  Bachelor of Technology in Computer Science Engineering — *July 2023*

Experience

- **AI Security & Governance Engineer** — Jan 2026 – present
  AT&T — *Plano, TX (Hybrid)*
  – Designed and secured enterprise AI governance workflows for GenAI and agentic AI systems, reducing AI use-case review and permit-to-operate approval time from **10–12 days to under 6 minutes** through automated agentic security review pipelines
  – Developed privacy-preserving tokenization and masking frameworks for enterprise networking data, enabling sensitive attributes such as IP addresses, VLANs, and infrastructure metadata to be securely transformed while preserving semantic context for AI and LLM-based systems.
  – Performed adversarial testing, architecture security reviews, and behavioral risk assessments on enterprise LLM systems prior to deployment, validating resilience against prompt injection, excessive agency, insecure tool use, and sensitive data exposure risks
  – Created enterprise AI security review frameworks, governance checklists, and runtime control requirements to standardize secure AI deployment practices aligned with zero-trust principles, AI risk management, and secure AI SDLC standards.
- **GenAI & Data Science Intern** — June 2025 – Dec 2025
  NOV Inc (National Oilwell Varco) — *Houston, TX (On-site)*
  – **Contact Us AI Automation & Email Responder:** Constructed complete automation system with AI engine that routes queries to appropriate personnel based on company content, website page context, user intent, and query location, eliminating manual forwarding and **saving 20+ hours of manual work weekly**. Formulated threat model and secure architecture for customer support automation system. Adhered to OWASP Top 10 standards, tracing risks throughout the AI Lifecycle phases, integrated OWASP Agentic AI Threat Modeling, defended against Prompt Injection (direct/indirect) and Jailbreak attacks. Achieved 0 critical vulnerabilities, 100% compliance, 96% response time processing 50+ emails daily (95% accuracy).
  – **Self-Improving MUD Report AI Automation (Accepted Paper):** Architected autonomous multi-agent GenAI system to accelerate PDF parsing (previously required 2 days per report). Orchestrated 3-agent system (vendor detection, prompt optimization, extraction) using Azure OpenAI (GPT-4o/5). Applied OWASP Top 10 for LLMs, identifying risks at all AI Lifecycle stages, executed OWASP Agentic AI Threat Modeling, defended against indirect Prompt Injection in pdf docs and Unbounded Consumption, applied NIST AI-RMF. Achieved 89-100% accuracy, 360x speed improvement (8 min vs 2 days), 0 security incidents.
  – **Secured an MCP Server for Expression Language Conversion:** Created a secure MCP server converting natural language to Expression Language(EL). Identified and mitigated critical MCP server vulnerabilities including command injection (RCE), weak authentication, rate limit, and tool poisoning. The project involved fine-tuning a Code-llama 8B model, mapping risks across the AI Lifecycle, and **enabling secure EL conversion for 5+ internal tools**.
  – Supported other data scientists and automation engineers to keep systems secure by consulting them on secure agentic coding and AI systems. Conducted security testing of their AI solutions and provided feedback
- **Research Assistant – AI & Cybersecurity** — Sep 2024 – May 2025
  University of Houston — *Houston, TX (On-site)*
  – Pioneered LIMA (first author) LLM-driven penetration testing framework (MCP-based), achieving 95% success rate, reducing testing time from 8+ hours to 15 minutes.
  – Engineered the PentestThinkingMCP server integrating reasoning agents with Nmap, Metasploit, and Burp Suite to enable vulnerability discovery across 50+ scenarios (90% accuracy). Applied OWASP Agentic AI Threat Modeling and secured the server with prompt-injection defenses, input/output validation, and rate-limiting controls. (Deployed Link: pentestthinkmcp)
  – Established quantitative benchmark: Claude 3.5 outperformed expert pentesters on 12/15 HTB boxes ($0.05/run, 95% cost reduction). Authored IEEE FMLDS 2025 paper on Secure AI Pentesting Frameworks
- **Associate Software Engineer** — March 2023 – February 2024
  Nagarro Software Pvt. Ltd.
  – Engineered scalable backend solutions (C#, .NET Core, SQL Server) with security controls, secure coding, input validation, SQL injection prevention, optimizing queries (30% improvement) across 25+ APIs.
  – Delivered secure REST APIs with JWT, RBAC, OWASP compliance, handling 1000+ daily requests (99.9% uptime), protecting against web vulnerabilities in 6+ applications.
  – Streamlined data analytics pipelines with security monitoring, reducing reporting by 40% (20 to 12 hours weekly).

Skills & Certifications

- **Programming & Scripting:** Python, C#, C++, SQL, Bash/Shell, PowerShell, JavaScript/TypeScript
- **Frameworks & Libraries:** FastAPI, Flask, .NET Core, Angular, LangChain, Hugging Face, Llama/Ollama
- **Security Tools:** Metasploit, Burp Suite, Nmap, Wireshark, Splunk, Cortex XSOAR, Docker, enum4linux, SAST/SCA Tools (Semgrep, GitLeaks, Trivy)
- **AI Security & Testing:** Prompt Injection & Jailbreak Defense, Adversarial AI Testing, Secure RAG, Data Poisoning Mitigation, Model Monitoring, LLMOps Security Controls
- **AI & Security Frameworks:** OWASP Top 10 for LLMs, Agentic AI Threat Modeling, AI Lifecycle Mapping, NIST AI-RMF, ISO 42001, Secure SDLC, AI Governance
- **Cloud & Infrastructure Security:** Azure Cloud Security (Key Vault, Managed Identities, Private Endpoints, Defender for Cloud), Azure OpenAI, Kubernetes, GitLab CI/CD
- **Penetration Testing Expertise:** Web App & Network Pentesting, Vulnerability Discovery, OWASP, Secure Code Review, Threat Modeling, AI Red Teaming
- **Certifications:** OWASP Top 10 for LLMs, Security+ (in progress), Azure 900, Azure AI-900, ISC2 CC, Fortinet NSE 1-3

Projects

- **PentestThinkingMCP - Autonomous Pen-Testing Server (IEEE Paper)** | *MCP, Red Team, AI Agent* | Live Link
  - Engineered MCP server enabling LLM-driven attack chains (Metasploit, Nmap), compromising HTB "Lame" in 3 minutes ($0.03, 95% cost reduction), demonstrating autonomous exploitation across 20+ scenarios.
  - Conceptualized modular AI agent framework (beam search, MCTS) for autonomous reconnaissance, privilege escalation, post-exploitation, integrating Burp Suite, improving efficiency by 80% (8 to 1.6 hours).
  - Deployed automated vulnerability discovery workflows enabling comprehensive security assessments support.
- **Contact Us AI Automation Threat Modeling Report** | *AI Threat Modeling, OWASP Top 10 for LLMs* | Report Link
  - Authored comprehensive threat modeling report applying OWASP Top 10 for LLMs framework, mapping risks (LLM-01: Prompt Injection, LLM-06: Sensitive Information Disclosure, LLM-08: Excessive Agency) across AI lifecycle phases.
  - Applied OWASP Agentic AI Threat Modeling to analyze agent risks. Documented threat models aligned with NIST AI-RMF and ISO/IEC 42001 compliance.
  - Analyzed Prompt Injection types (direct/indirect) and Jailbreak vulnerabilities, providing defense mechanisms and validation strategies for production deployment.
- **Agentic Lean Embedding System for Vulnerability Discovery** | *Python, LangChain, GitLeaks* | **Active Research Project**
  - Developing agentic pipeline combining lean embeddings for vulnerability discovery.
  - Building semantic indexing system for software ecosystems, enabling OS system-wide vulnerability discovery 90% faster, and reduced false positives better efficiency.
  - Implementing cross-source correlation and provenance tracking, identifying 500+ secrets, 200+ insecure API patterns, 150+ risky settings with human-in-the-loop triage.
- **Cybersecurity Awareness Platform** | *Python, JavaScript, Flask* | sctool.web.app
  - Launched interactive platform with 30+ penetration testing tools (Metasploit, Burp Suite), serving 1,000+ learners.
  - Integrated attack simulations, automated scoring, and progress tracking, enabling 500+ users to learn secure practices.
  - Curated secure coding workshops and training modules (Python, C#), teaching 200+ developers to remediate security flaws.
- **Tax Application Threat Modeling** | *Secure Software Engineering, Threat Modeling, OWASP* | Report Link
  - Conducted comprehensive threat modeling for tax application system, identifying and mitigating vulnerabilities across architecture, data flow, and trust boundaries.
  - Performed domain analysis and security pattern assessment, mapping threats to OWASP Top 10 categories with mitigations.
  - Applied secure software engineering principles, documenting threat models, attack trees, and security controls for comprehensive risk coverage.

Research

- **Saleem, M. I.** (First Author), et al. "*LIMA: Leveraging Large Language Models and MCP Servers for Initial Machine Access.*" *IEEE FMLDS*, 2025 **(Published)** Paper Link
- NOV GenAI Team, **Saleem, M. I.** (Second Author). "*Self-Improving GenAI Agent for Fully Automated Report Parsing in Enterprise Environments.*" **SPE**, 2025 **(Published)** Paper Link
- **Saleem, M. I.** (Lead Researcher). "*Agentic Lean Embedding System for Vulnerability Discovery.*" **Active Research**.
- **Saleem, M. I., Banerjee, Tania** "*EncoderThinkingMCP: Guided Encoder-Decoder Model Development via MCP.*" AI-driven MCP server providing structured workflows, reasoning, and multi-framework support for encoder-decoder architectures. *IEEE Southwest*, 2026 (In experiment phase).